Exploring LLM-based Chatbot for Language Learning and Cultivation of Growth Mindset

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ABSTRACT

Growth mindset, the belief in enhancing abilities through effort and dedication, is commonly applied in classroom learning. The Mindset Theory Scale suggests it can be measured by "effort" and "belief in improvement", while fixed mindset is indicated by "procrastination" and "immutability of belief". In today's interconnected world, the need for proficiency in multiple languages has grown. However, mastering different languages poses several challenges, including motivational challenges and fear of failure. To address this issue, our study explores the use of an adaptable Large Language Model (LLM) based chatbot to foster growth mindset and aid in language learning. In a user study with this novel chatbot system, we evaluated the impact of a growth mindset teaching style on new language learning and user perception. Our initial findings show that users are more comfortable, confident, and interested in interacting with the growth mindset chatbot compared to the fixed mindset chatbot.

CCS CONCEPTS

• **Human-centered computing** → **Interactive systems and tools**; • **Computing methodologies** → *Artificial intelligence.*

KEYWORDS

Growth Mindset, Chatbot, LLM, Online Language Learning, Grit

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1 INTRODUCTION

1.1 Background

In an increasingly globalized world where more than half of people are bilingual, fluency in multiple languages is in high demand [2]. Despite this, many experience difficulties in mastering a new language due to motivational challenges, cultural nuances, and fear of failure. To tackle these challenges, we have developed an AI-driven chatbot designed to provide language instruction while utilizing a growth mindset. This approach is intended to transcend conventional teaching, alleviating language-learning apprehensions, amplifying innate motivation, advocating for self-efficacy, and triggering positive cognitive and emotional changes. While previous work has analyzed the use of Large Language Models in language learning, as well as the use of new technologies (chatbots and robots) in promoting a growth mindset within users, no research has analyzed the effects of an LLM-based chatbot designed to take a growth mindset-based approach to language instruction[4–7].

In creating our chatbot, we based the definition of a growth mindset and fixed mindset on the Mindset Theory Scale (MTS)[8]. The MTS measurement tool was validated for use in determining fixed mindset and growth mindset levels. As defined by the MTS, the two components of fixed mindset are procrastination and the Immutability of Belief. The two components of growth mindset are "effort" and "belief in improvement". To evaluate the mindset change of participants, we created questionnaires based on the MTS.

1.2 Overview

While there is existing literature focusing on the effect of chatbots on language learning and literature analyzing the efficacy of robots and chatbots in fostering a growth mindset, there is a lack of research exploring the potential of Large Language Models in nurturing a growth mindset within language learners. To address this gap, we have developed a web platform that incorporates an AI-driven chatbot with a growth mindset. The platform guides users through adaptive and personalized language learning. To this end, we have formulated two research questions:

(1) How can an AI-driven chatbot utilizing a growth mindset-based approach to instruction affect users' mindsets?
(2) Can an AI-driven chatbot utilizing a growth mindset-based approach enhance users' language learning capabilities?

2 RELATED WORK

Previous work has examined the impact of a growth mindset versus a fixed mindset within technological interactions. One study explores the use of a social robot to foster a growth mindset within children [6]. Results showed that children interacting with the growth mindset robot both self-reported having a stronger mindset and tried harder during challenging tasks than the children interacting with a neutral robot.

Several studies have explored the use of chatbots within language learning. One such study analyzes the use of a chatbot within a language learning app for ESL learners [7]. Although the chatbot was difficult to use, users did interact positively with the chatbot.

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Based on the results of this study, the authors suggest that chatbots within an educational app should have a "simple and friendly design" and that they should "provide detailed instructions." In our study, we incorporated these ideas to maximize the instructional capabilities of our chatbot. However, while the chatbot is the main feature of the app, this study did not analyze user interactions with the bot. Another similar study highlights the potential of AR glasses combined with ChatGPT to improve oral language skills [4]. Researchers created a system, VisionARy, that scans and recognizes the users' surroundings "recognizes the users' surroundings and offers oral correction." This paper concludes that "VisionARy leads to statistically significant improvements in interface autonomy, task competence, aesthetic appeal, and reward."

Separate research underlined the importance of promoting positivity in digital learning by merging project-based learning and a growth mindset [5]. This study concluded that growth mindset instruction was effective within education. The authors of the paper notably recommended for improvement that "Artificial intelligence technology should be also added to improve students' positive thinking Repository." Within our study, we aim to analyze the effectiveness of AI technology in promoting a growth mindset and positive thinking.

Currently, there is a lack of research exploring the potential of Large Language Models to nurture a growth mindset in learners. We aim to bridge this gap by creating an AI-driven chatbot designed to provide language instruction while utilizing a growth mindset. To do this, we created two chatbots, one with a growth mindset and the other with a fixed mindset, to analyze the effects of chatbot mindset on AI-based language learning.

3 INTERFACE DESIGN

We created two different web-based chatbots: one with a fixed mindset and the other with a growth mindset using ChatGPT's API. We chose to utilize the ChatGPT API because of its customizability, enabling us to design both system prompts and the web interface, in addition to its adaptability to user questions. Each chatbot was prompted to teach users ten different Latin words throughout their conversations with the chatbots.

In the first session, the chatbots were instructed to teach users the words: Animus, Aqua, Arbor, Flora, Herba, Luna, Natura, Sol, Terra, and Ventus. In the second session, the chatbots were instructed to teach users the words: Amicitia, Audax, Clementia, Felicitas, Fides, Fortitudo, Gravitas, Pulcher, Sapientia, and Tempus. To program each chatbot with a different mindset, we modified the role of the system within the conversation history parameters.

3.1 Prompt Design

To generate the system prompts, we began by compiling a list of characteristics associated with both a fixed and growth mindset. Individuals with a fixed mindset generally believe that their talents and skills are static and cannot be improved. In contrast, those with a growth mindset believe that their abilities can be developed through hard work and dedication [1]. We incorporated these characteristics into the system prompts to shape the chatbot's teaching style and attitude. Additionally, we provided instructions to the chatbot on formatting its responses, how to respond to user mistakes and questions, and the specific list of words it would be teaching the user.

The growth mindset chatbot was given this prompt: "You are a supportive teacher who believes in a growth mindset. You want to adapt to the user's language learning needs and support them in their language journey. You're not repetitive but want to check-in frequently and be available to help." In contrast, the fixed mindset chatbot was given this prompt: "You are a teacher who is obsessed with rules and proper grammar. You are pedantic. You will point out when the user makes mistakes, and comment on when they are wrong. Also keep track of how many words learned and how many mistakes made so far and let the user know occasionally."

4 METHODOLOGY

The experiment was designed to gain insights into user perceptions of the chatbots, including their attitudes toward the agents and their behavior while interacting with them. The study is a between-subject design, meaning that each participant interacts with only one of the chatbots with either a growth mindset or fixed mindset teaching style. All of the studies were conducted remotely through Zoom. We chose Latin as the language our chatbot would teach because it is not commonly studied in college; therefore, all our participants would have the same amount of prior experience.

4.1 Participants

Participants were recruited from undergraduate students at a liberal arts college in the United States using convenience sampling via email, text, and shared forums. A total of 13 students signed up for the study, and 12 (N=12) completed both research sessions. Participants were primarily females between the ages of 18-21, all pursuing an undergraduate degree. These participants were randomly, but evenly, assigned either to the growth mindset or fixed mindset chatbot.

4.2 Questionnaire Design

4.2.1 Pre-Study Questionnaire. The pre-study questionnaire, consisting of 10 questions, assessed participants' grit level and anxiety in foreign language learning. We utilized Angela Duckworth's Grit scale, a five-point Likert scale, to measure perseverance and self-image [1]. Additionally, we employed the Foreign Language Classroom Anxiety Scale (FLCAS) using a similar Likert scale to gauge comfort in learning a foreign language [3]. We used these scales as they are easy for individuals to complete individually and measure an individual's perseverance and self-image

4.2.2 Post-Study Questionnaire. The post-study questionnaire, comprised of 15 questions, documented participants' reactions to the chatbot and measured changes in their perseverance and feelings toward foreign language learning. We included open-response questions prompting users to reflect on their interaction with the chatbot, identify any notable interactions, and provide adjectives to describe the chatbot. Questions from Duckworth's Grit scale and FLCAS were also included for comparison with participant's initial self-assessment.

4.3 Word Search Puzzle

To measure their increased learning through the chatbot interaction, we employed word search puzzles. We selected word puzzles as a means to gauge learning due to their ease of individual completion, visual engagement, and quick generation. Two word puzzles were created: one with "easier" vocabulary words and another with "harder" vocabulary words. The easier vocabulary words were all themed around nature and ranged from six to eight letters in length. The harder vocabulary words were themed around concepts and were approximately 10-12 letters long. Participants were not provided with a word bank when asked to complete the word puzzle.

4.4 Protocol

Throughout this procedure, we recorded the time each participant took to complete the puzzle, the number of words they guessed correctly and incorrectly while completing word puzzles, the time it took to interface with the chatbot, and their chat logs. Tasks 1-3 were repeated for Session 2 with the more challenging vocabulary list of Latin words. Additionally, after the pre-questionnaire, users were shown the first session word puzzle to gauge their recall of the previous lesson.

Task 1: Screening Questionnaire / Pre-Session Word Puzzle

At the start of the study, participants were asked to complete a screening questionnaire to provide demographic information. Users were also queried about their familiarity with chatbots, AI, and language fluency. Following this, participants completed our standardized pre-study questionnaire and were tasked with completing the first word puzzle. They were informed that they had up to 5 minutes to find 10 Latin vocabulary words, but could end earlier.

Task 2: Chatbot Interaction

Participants were then introduced to our chatbot, LinguaGemma, which was locally hosted on the researcher's computer conducting the study. Once the interface loaded, they were directed to the input box, where they could begin their conversation by typing a message to the chatbot. Users were informed that the chatbot would guide them through learning 10 Latin vocabulary words, and they would have up to 20 minutes to interact with the bot. Additionally, they were advised that the chatbot should be reactive and adaptive to their questions or comments.

Task 3: Post-Questionnaire / Post-Session Word Puzzle

After users had learned all 10 vocabulary words, or 20 minutes had passed, they were provided with our post-study questionnaire. Lastly, participants were shown the same puzzle from the start of the session and, once again, given up to five minutes to find the 10 Latin vocab words they had just learned.

5 DATA ANALYSIS

To understand how participant behavior varied between the agents, we (i) calculated growth mindset scores based on questionnaires (ii) used thematic analysis on the chatbot interaction (iii) compared users' overall comfort level and conducted semantic analysis based on adjectives reported. In addition, we used (iv) word search puzzles to measure language learning throughout the interaction. For our preliminary study, we used non-parametric Mann-Whitney's U test to find distributional differences between the two chatbot groups. To check assumptions for our hypothesis test to compare two different groups of participants, we ran Shapiro-Wilk (S-W) test to check for normality and Levene's test to check for equal variance. As a result, we were not able to conclude that our datasets meet normality assumption, but the two groups of chatbot approximately had within and between equal variances. Therefore, we used non-parametric Mann-Whitney's U test to compare the two different styles of chatbot in the following analysis.

5.1 Growth Mindset Change

We first analyzed questionnaires to calculate growth mindset scores and detect distributional differences for each growth mindset question. Four questions, featured in our post-study survey, related to the "effort" and "belief in improvement" components of growth mindset. They were rated on a likert-scale of five, with a maximum score of 20. Each question was examined for distributional differences between chatbot groups. The null hypothesis assumed no difference between treatment groups, while the alternative hypothesis suggested a difference.

5.2 Thematic Analysis on Chatbot Interaction

To check if we can also measure the participants' change between the two chatbots and throughout the session by completing a thematic analysis on their interaction. Specifically, we evaluated the number of times participants asked the chatbot for more examples in order to understand how engaged the user is with the chatbot's lesson; asked unrelated questions or attempts to change the chatbot's teaching style in order to understand when the user is distracted or disinterested in language learning; expressed familiarity with a word to assess a users previous language knowledge; and expressed discouragement in order to assess the users confidence language learning. We conducted this additional thematic analysis to suggest alternative methods to understand the interaction using the chatbot conversation rather than questionnaires.

5.3 Overall Attitude towards Language Learning with Chatbot

In addition to their change in growth mindset and language acquisition, we completed an analysis to find participants' general attitude toward using chatbots for language learning, comfort level, and perceptions of chatbot teaching styles. We included questions in the final questionnaire that gauged overall comfort with the chatbot on a Likert scale, and users' descriptions of teaching styles were analyzed and sorted by positive, neutral, or negative sentiments. Moreover, we also asked users "What are three adjectives you would use to describe the chatbot's teaching style?"

5.4 Vocabulary Learning and Retention

For our second research question, whether growth mindset enhances language learning, we looked into five word search puzzles to measure their language learned and retained. Using the initial word search puzzle and post-interaction word search puzzle for each two sessions and one retention word search puzzle at the beginning of the second session, we were able to quantify newly learned words throughout the interaction.

6 RESULTS

6.1 Calculating Growth Mindset Score

Overall, we discovered that participants who interacted with growth mindset chatbot exhibited higher growth mindset scores compared with those who interacted with the fixed mindset chatbot. Before chatbot interaction, the growth mindset chatbot group had an average score of 17, and the fixed mindset chatbot group showed an average score of 15. After two rounds of interaction, the growth mindset group increased to an average of 20, and the Ffxed mindset group decreased to an average of 13. The growth mindset group's score rose by 3 points, while the fixed mindset group's score dropped by 2 points on average. This indicates successful creation of two chatbots promoting different growth mindset levels.

To understand the growth mindset change, we examined questions related to growth mindset scores for each category. Mann-Whitney U tests at a 0.05 significance level revealed a significant difference in resistance to discouragement between the two groups. While the users' resistance to discouragement changed significantly after chatbot interaction, no significant differences were observed in other growth mindset categories, despite growth mindset groups consistently having higher average scores in each category.

6.2 Thematic Analysis on Chatbot Interaction

As seen in table 1, thematic analysis also revealed noteworthy interactions and comparisons across different chatbot styles throughout the session. Our preliminary results indicated an increase in discouragement in the fixed mindset group compared to the growth mindset group. As we cannot conclude that each notable action is indicative of growth mindset without proven results on their association, we have not concluded further about other categories. However, we were able to see notable changes from Session 1 to Session 2 for both groups. Interestingly, in the first session, fixed mindset chatbot participants, on average, exhibited more instances of asking for examples, posing general language questions, attempting to change teaching style, and indicating familiarity with words compared to the growth mindset chatbot users. However, in the second session, fixed mindset chatbot users experienced a significant decrease in these occurrences, while growth mindset chatbot users increased their instances of asking for examples, general language questions, attempts to change teaching style, and familiarity with words.

6.3 Overall Attitude towards Language Learning with Chatbot

To understand participants' overall perception of the chatbot, we asked about their general comfort level during interactions. Our null hypothesis assumed no difference in comfort based on the chatbot's teaching style. With a p-value of $p=0.020$, we found sufficient evidence to conclude that users felt more comfortable with the growth mindset chatbot. Growth mindset chatbot users reported a median of 4.5 and a mean of 4.3, while fixed mindset users reported a median of 3 and a mean of 2.8. This indicates a 34% higher comfort level with the growth mindset chatbot. Overall, the growth mindset chatbot fostered a more comfortable and language-learning-friendly interaction compared to the fixed mindset chatbot. Additionally, semantic analysis revealed that users interacting with the growth mindset chatbot used more positive descriptions and adjectives to characterize its teaching style than users with the fixed mindset chatbot.

6.4 Vocabulary Learning and Retention

In language learning, we were able to find differences between the two chatbots. We were able to see that 14.58% more words were acquired for those interacting with growth mindset chatbot compared to the fixed mindset chatbot in the first session, and 25.0% more words were learned in even more challenging scenarios.

Moreover, we looked into the number of words retrieved to measure their language retention. Growth mindset chatbot participants retained 37.8% more words compared to those with fixed mindset chatbot. Using the Mann-Whitney U Test, with a p-value of $p=0.029$, we were able to reject the null hypothesis that there is no difference between the two groups that interacted with different chatbots. We have strong evidence to conclude that the growth mindset chatbot helped learners retain significantly more words than the fixed mindset chatbot.

7 DISCUSSION AND LIMITATIONS

Throughout our study, we developed an accessible language-learning LLM-based chatbot showcasing two learning styles: growth mindset and fixed mindset. Additionally, we were among the first to quantify chatbot interactions, calculate the growth mindset score in language learning, and reveal significant differences in the expression of discouragement among participants who interacted with two different chatbots. The current findings suggest a correlation between interaction with the growth mindset chatbot and increases in the number of words learned and retained. Moreover, we identified a correlation between the growth mindset chatbot and increases in growth mindset, as defined by the Mindset Theory Scale (MST) [8].

Areas for further research and development include the need for more participants to gather additional data points and evaluate the impact of the growth mindset chatbot. Our study was limited by a small participant pool (N=12), prompting the call for more detailed studies to investigate the long-term impact of growth mindset-based AI chatbots on language learning. The small sample size also hindered our ability to claim statistical significance. Also, recruiting a more diverse set of participants in terms of age, education, and language learning experience is crucial for broader applicability. Addressing these limitations may reveal larger differences in growth mindset score changes, learning capabilities, and lead to more meaningful conclusions.

8 CONCLUSION AND FUTURE WORKS

With the rise of globalization and automation, investigating how AI technology can be used to support language learning is more critical than ever [2]. Applying a growth mindset to this learning could be incredibly productive and influential in new language learning and retention.

In this paper, we investigated the impact of chatbots with growth and fixed mindsets on Latin vocabulary learning and their influence on user growth mindsets. Throughout our study, we found

Table 1: Thematic Analysis of User's Chatbot Interactions (Average Number of Responses Per Category)

	Times asked for more examples	Times asked unrelated questions	Times tried to change teaching style	Times indicated familiarity with a word	Times expressed discouragement
Session 1: Growth	1	0	0	3.67	0
Session 1: Fixed	1.67	0.67	2	7.33	3.33
Session 2: Growth	0.75	0.75	0.75	1	1
Session 2: Fixed	0.67	0	0	0	1

Table 2: Thematic Analysis of User's Chatbot Interactions (Growth Mindset Chatbot vs Fixed Mindset Chatbot)

	Average Number of Positive Adjectives	Average Number of Neutral Adjectives	Average Number of Negative Adjectives
Session 1: Growth	2.33	0.33	0.17
Session 1: Fixed	0.75	1	1
Session 2: Growth	1.75	0	1.25
Session 2: Fixed	0.75	1	1.25

Table 3: Average number of words learned and retrieved (Growth Mindset Chatbot vs Fixed Mindset Chatbot)

	Session 1: Newly Learned Words		Session 2: Retrieved Words		Session 2: Newly Learned Words	
	Mean	Median	Mean	Median	Mean	Median
Growth	8	8	8.7	9	7.5	7.5
Fixed	6.8	6.5	6.7	6.5	4.67	4.5

significant differences in the distribution of resilience to discouragement between two chatbots, with the growth mindset chatbot exhibiting higher mean and median scores. Users reported a greater comfort level when interacting with the growth chatbot as opposed to the fixed chatbot. Results from the questionnaire highlighted increased confidence in overcoming setbacks for growth mindset chatbot users, and word puzzle outcomes demonstrated enhanced word recall. Thematic and sentiment analysis reflected that growth mindset chatbot participants predominantly used positive adjectives to characterize their interactions, while fixed mindset chatbot participants leaned towards negative and neutral descriptors.

Our research holds particular significance as it is one of the first to directly measure growth mindset scores through chatbot interfacing. Our development of a versatile language learning chatbot, demonstrated within just two rounds of study, showcases its potential adaptability for diverse applications and its seamless adjustment to various tasks and contexts.

In future work, we would like to increase the number of participants, the duration of this study, as well as the complexity of Latin vocabulary taught. While we measured interaction time, we limited the maximum amount of time participants could interact with the chatbot, creating censored data. Looking forward, we intend to give unlimited time for users to converse. As we found that most participants using the growth chatbot felt "more comfortable" during their interaction, we hypothesize that unlimited and untimed access to the chatbot could lead to an increase in interaction time.

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